

Beyond the Needle: Stress-Testing Long-Context Intelligence

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Abstract—While modern Large Language Models (LLMs) have significantly larger context windows that can handle millions of tokens, this does not guarantee reliable comprehension. State-of-the-art models still struggle with realistic long-context tasks, which shows a clear need to identify where and why these models fail. This project suggests a systematic, diagnostic evaluation framework to explore long-context failures using an improved version of the LongBench v2 benchmark. Instead of relying on simple retrieval tests and overall accuracy scores, we introduce controlled perturbations to pinpoint specific weaknesses. Our method uses context length scaling to measure accuracy loss, needle position adjustments to reveal positional retrieval biases, and different prompting strategies to separate retrieval issues from reasoning limitations. By looking at these aspects, we aim to create a detailed error taxonomy that sorts failures into retrieval failures, reasoning limitations, and instruction non-compliance. Ultimately, this work shifts long-context evaluation from fixed benchmarking to dynamic analysis, giving us a deeper understanding of how LLMs decline in performance and the trade-offs between accuracy and efficiency.

I. INTRODUCTION

Large Language Models (LLMs) have advanced to the point where they can process millions of tokens in a single forward pass, substantially expanding their context windows. However, raw capacity does not imply reliable comprehension. As context windows grow, a critical and underexplored question emerges: where, precisely, do these models break down—and why?

This project investigates long-context failures using LongBench v2 as the primary evaluation framework [1]. LongBench v2 is designed to assess LLM performance on realistic tasks requiring deep reasoning over extended contexts. The benchmark comprises 503 multiple-choice questions with contexts ranging from 8,000 to 2 million words, spanning six major categories: single- and multi-document question answering, long-form in-context learning, dialogue history understanding, code repository analysis, and structured data reasoning [1]. Despite the scale and diversity of tasks, the top-performing model achieves only 50.1% accuracy—only marginally comparable to the 53.7% human expert baseline—highlighting a persistent performance gap in long-context reasoning [1].

The primary objective of this work is diagnostic evaluation. Specifically, we aim to determine whether model failures stem from:

- **Retrieval failure** — inability to locate relevant information in long contexts;

- **Reasoning failure** — inability to logically combine retrieved evidence;
- **Instruction non-compliance** — failure to adhere to task constraints despite sufficient information.

The input consists of long-form documents paired with a question, and the desired output is a correct answer supplemented by structured error attribution in cases of failure.

Standard evaluations such as the classic Needle-in-a-Haystack (NIAH) tests often assess simple lexical retrieval and can provide an overly optimistic estimate of long-context capabilities [2]. While modern LLMs frequently achieve near-perfect performance on synthetic single-span retrieval tasks, performance declines substantially in more realistic scenarios involving information sparsity, distractors, and multi-hop reasoning [2], [3]. Research further demonstrates that retrieval success is highly sensitive to the relative size and placement of relevant spans within the context [3]. More recent work emphasizes that real-world long-context evaluation must move beyond simple extraction toward structured, heterogeneous, and agentic reasoning tasks [4].

Aggregate accuracy scores provide limited insight into the structure of these failures. This project therefore aims to develop a fine-grained error taxonomy by systematically varying context length, manipulating evidence placement, and comparing prompting strategies such as extract-then-answer versus direct-answer prompting. The resulting accuracy-versus-length curves and error breakdowns provide a more diagnostic view of the limitations of long-context modeling.

II. RELATED WORK

Recent research has introduced large-scale long-context benchmarks to evaluate large language models (LLMs) on inputs spanning tens of thousands to millions of tokens. While early results suggested strong long-context capabilities, more recent studies reveal substantial limitations. Prior work consistently identifies three major failure modes: **(1) retrieval brittleness**, **(2) multi-hop reasoning limitations**, and **(3) instruction non-compliance**.

A. Large-Scale Long-Context Benchmarks

Several benchmarks systematically evaluate long-context performance. **LongBench v2** includes 503 multi-task evaluations with contexts up to 2M words and reports that even the best model achieves only 50.1% accuracy, compared to 53.7% for human experts [1]. Other prominent suites include

ZeroSCROLLS [5], **L-Eval** [6], **BAMBOO** [7], **LooGLE** [8], **Bench** [9], **RULER** [10], and **HELMET** [11].

However, many benchmarks emphasize extractive retrieval or rely on surface-overlap metrics (e.g., ROUGE, F1), which can overestimate true comprehension [9], [10]. Simple tests such as **Needle-in-a-Haystack (NIAH)** often yield near-perfect scores [9], creating the impression that long-context understanding is largely solved. More realistic tasks requiring aggregation and reasoning expose substantially greater weaknesses.

B. Retrieval-Centric “Needle” Evaluations

Although standard NIAH tests are easily solved [9], retrieval robustness declines under more challenging settings. **Lost in the Haystack** shows that performance drops sharply when the relevant span is very small relative to the context or positioned unfavorably [3]. **Context Rot** demonstrates that increasing irrelevant filler, especially semantically plausible distractors, degrades accuracy [2].

Benchmarks such as **RULER** further show that retrieval performance deteriorates as context length and task complexity grow [10]. These findings suggest that naïve extraction benchmarks mask brittle retrieval behavior.

C. Multi-Hop and Dense Reasoning over Long Contexts

Beyond single-span lookup, models struggle with tasks requiring aggregation of dispersed evidence. **NeedleBench** distinguishes between sparse retrieval and dense reasoning tasks, showing strong performance on isolated lookup but substantial drops on multi-step inference (e.g., ancestral tracing) [12]. Similarly, **HELMET** emphasizes comprehensive evaluation of reasoning across long contexts [11].

Overall, aggregating distributed evidence is significantly more difficult than direct extraction.

D. Prompting and Decomposition Strategies

Structured prompting methods aim to mitigate long-context failures. **Path-of-Thoughts** extracts relational graphs before reasoning and improves controlled multi-hop tasks [13]. However, extract-then-answer approaches can amplify errors when retrieval fails. For example, evaluation on scientific QA found that extraction mistakes significantly reduced overall accuracy compared to direct-answer prompting [14].

Thus, decomposition helps only when evidence is reliably identified.

E. Instruction-Following Failures

Beyond retrieval and reasoning, LLMs also exhibit **instruction non-compliance**. The “instruction gap” describes cases where models derive correct answers but violate formatting or procedural constraints [15]. This represents a distinct failure mode in long-context settings.

Prior work identifies three primary sources of long-context failure:

- Retrieval brittleness, particularly under realistic or distractor-heavy conditions [2], [3];

- Reasoning limitations in aggregating dispersed evidence across long contexts [10], [12];
- Instruction non-compliance, even when retrieval and reasoning succeed [15].

Our work builds on these insights to provide a unified and systematic analysis that disentangles these failure modes through controlled manipulation of context length, evidence distribution, and prompting strategies.

III. DATASET

We will base our study on LongBench v2 [1], a realistic long-context benchmark consisting of 503 challenging multiple-choice questions with context lengths ranging from 8K to 2M words across six categories: single-document QA, multi-document QA, long in-context learning, dialogue history understanding, code repository reasoning, and structured data understanding. The benchmark is specifically designed to evaluate deep reasoning over extended inputs and provides reliable evaluation through its standardized multiple-choice format.

To systematically characterize long-context failure modes, we will extend LongBench v2 with controlled diagnostic variants rather than relying solely on the original benchmark distribution.

First, we will construct context-scaled variants by truncating each task’s context to predefined token budgets (e.g., 8K, 32K, 64K, 128K). This enables us to measure performance degradation as a function of context length and estimate accuracy length slope trends.

Second, for tasks where applicable, we will introduce needle-position perturbations by relocating answer-relevant spans to different positions within the context (beginning, middle, end). This controlled manipulation allows us to isolate retrieval sensitivity from reasoning complexity.

Third, we will create a manually annotated subset of incorrect predictions to categorize errors into retrieval failures, reasoning failures, and instruction compliance failures. This taxonomy will support fine-grained analysis beyond aggregate accuracy metrics.

By augmenting LongBench v2 with these controlled modifications, we convert a static benchmark into a diagnostic framework for analyzing long-context degradation patterns in modern language models.

IV. METHODOLOGY

To systematically characterize long-context failure modes, we conduct controlled evaluations across multiple experimental dimensions: context length, positional placement of relevant information, and prompting strategy.

A. Model Selection

We evaluate several open-source language models supporting extended context windows ($\geq 32K$ tokens). Models are selected to represent diverse architectural strategies for long-context handling. All experiments use deterministic decoding (temperature = 0) to ensure reproducibility for multiple-choice evaluation.

B. Context Length Scaling

To analyze performance degradation as a function of input length, we construct truncated variants of each LongBench v2 task at predefined token budgets (e.g., 8K, 32K, 64K, 128K), subject to model capacity constraints. This enables estimation of accuracy-versus-length curves and quantitative degradation slopes.

C. Needle Position Perturbation

For QA-style tasks, we identify answer-relevant spans and systematically relocate them to different positions within the context (beginning, middle, and end). This manipulation isolates retrieval sensitivity from reasoning complexity and allows measurement of positional bias.

D. Prompting Strategy Comparison

We compare two prompting approaches:

- **Direct Answer:** The model receives the full context and produces an answer directly.
- **Extract-Then-Answer:** The model is instructed to first extract relevant spans and then answer using only the extracted information.

This comparison enables separation of retrieval limitations from reasoning failures.

E. Evaluation Metrics and Error Analysis

Performance is measured using accuracy per task category and overall accuracy. We further compute accuracy-versus-length trends and estimate degradation slopes. In addition, we record latency and token usage to assess efficiency tradeoffs.

A manually annotated subset of incorrect predictions is categorized into retrieval failures, reasoning failures, and instruction non-compliance. This structured error taxonomy supports fine-grained analysis of degradation patterns.

Confidence intervals will be estimated using bootstrap re-sampling.

V. EVALUATION PLAN

A. Evaluation Objectives

The goal of our evaluation is not only to assess the performance of the model on LongBench v2, but also, more importantly, to identify the reasons for the failure of the model on the long-context test cases. To be specific, we aim to quantify how the performance of the model changes as the context length increases, identify whether the performance degradation is caused by failure of retrieval, reasoning, or instruction non-compliance, as well as how prompting strategies affect different types of failure. We also aim to evaluate the trade-off between accuracy and computation for long-context inference. By introducing perturbations, the evaluation framework changes LongBench v2 from a static evaluation benchmark to a dynamic analysis framework for understanding the behavior of the model on long-context test cases.

B. Experimental Setup

1) *Models:* The experiments will include a selection of various open or open-weight instruction models that differ in their parameters and capacity of their context window. For all models, their parameters' size, their capacity for a context window, their tokenizer, their decoding scheme, and their environment will be noted. In all cases where not otherwise stated, a temperature of zero will be used for decoding to ensure deterministic output.

C. Controlled Diagnostic Variants

1) *Context Length Scaling:* To measure robustness with respect to context length, we will create variants of LongBench v2 instances with a specified token limit. In particular, we will create variants with context lengths of 8K tokens, 32K tokens, 64K tokens, and 128K tokens. We will also include additional context lengths if computationally feasible. The full question and answer options will remain unchanged, and only the context will be truncated.

This controlled scaling allows us to measure model performance as a function of context length. We will compute accuracy at each context length, plot accuracy as a function of log context length, and estimate an accuracy degradation slope as a function of context length via linear regression. This single value captures model robustness to context length.

2) *Needle-Position Perturbation:* For tasks where answer-relevant spans can be identified, controlled needle position perturbations are introduced. In these variants, the identified answer-relevant span is moved to the beginning, middle, or end of the context. This tests for retrieval sensitivity to positional information and long-range attention effects.

If accuracy varies significantly with the position of the answer-relevant span, it suggests retrieval sensitivity. Otherwise, if accuracy remains constant despite the position of the answer-relevant span, it suggests that the model has limitations in reasoning.

3) *Prompting Strategies:* To further differentiate the retrieval and reasoning failures, we also explore different prompting strategies. In the Direct Answer Prompting (DA) strategy, the model is provided the entire context and asked to directly retrieve the answer choice. In the Extract-Then-Answer Prompting (E2A) strategy, the model is first prompted to extract the relevant evidence from the context, and then it is asked to retrieve the final answer using only the extracted evidence. This will allow us to see if the performance improves when retrieval is supported by the different strategies. If E2A significantly improves the accuracy, it indicates that retrieval is the main source of the failures. If the performance remains poor even when the extraction is successful, it indicates that the failures are likely due to reasoning.

All the prompt strategies will be standardized to facilitate reproducibility.

D. Evaluation Metrics

1) *Accuracy:* The primary metric for performance is the notion of accuracy. We will report overall accuracy as well

as accuracy by task category, context length bucket, needle position, and prompting strategy. All of our reported accuracy metrics will come with 95% confidence intervals to account for statistical uncertainty.

2) *Accuracy-Length Degradation*: To measure robustness to context growth, we apply a linear regression of accuracy versus log context length for each model and category. The degradation slope represents how fast the performance of the model decreases when the context grows. This metric allows for easy comparison of robustness to long contexts for different models and prompting methods.

3) *Latency and Token Efficiency*: Another aspect we want to evaluate, in addition to accuracy, is computational efficiency. For this purpose, we track the number of input tokens, number of output tokens, total tokens, and the wall-clock time taken for inference on each example, for every evaluation run. This will help us study the accuracy vs. token cost and accuracy vs. latency curves to assess the efficiency of various models and prompting methods in long-context configurations.

E. Error Taxonomy and Attribution

Aggregate accuracy alone does not reveal why failures occur. To address this limitation, we introduce structured error labeling.

1) *Error Categories*: These incorrect predictions will be classified into three primary failure types: retrieval failures, reasoning failures, and instruction non-compliance. In addition, hallucination will be recorded as a secondary diagnostic attribute, defined as answers that rely on information not supported by the provided context.

2) *Annotation Procedure*: Error attribution will occur via a two-stage process. In the first stage, automatic heuristics will be used to identify evidence overlap (when possible), format errors, and answers without support. In the second stage, a stratified sample of incorrect predictions will be manually annotated. The stratified sampling will be used to cover a range of context lengths, task types, and prompting strategies. Two annotators will annotate each incorrect prediction, and any disagreements will be resolved via adjudication. Inter-annotator agreement will be measured via Cohen’s κ . The results will include error distributions by category and context lengths.

F. Statistical Analysis

In order to ensure the robustness of our conclusions, we will use bootstrap resampling to compute confidence intervals for all key metrics. McNemar’s test will be used for paired model comparison. When performing multiple model comparisons, corrections for these tests will be applied appropriately. In addition to statistical significance, we will also use effect sizes to give a clear picture of the practical significance of our results.

G. Reporting and Visualization

Results will be visualized by providing accuracy vs. context length curves, degradation rate comparisons, needle position

sensitivity heatmaps, error type distributions, and latency vs. accuracy tradeoff studies. Additionally, qualitative results will be given to describe typical characteristics of retrieval, reasoning, and instruction failures. All these visualizations will offer both quantitative and qualitative results on how well these models perform with longer context.

H. Reproducibility

In order to ensure reproducibility, random seeds will be fixed where necessary, and all prompts and decoding parameters will be logged. The hardware configurations will be provided in great detail. The evaluation scripts and modified benchmark variants will be released to aid in the reproduction of the results.

REFERENCES

- [1] Y. Bai *et al.*, “LongBench v2: Towards deeper understanding and reasoning on realistic long-context multitasks,” arXiv:2412.15204, Dec. 2024. [Online]. Available: <https://arxiv.org/abs/2412.15204>
- [2] K. Hong, A. Troynikov, and J. Huber, “Context Rot: How increasing input tokens impacts LLM performance,” Tech. Rep., Chroma, Jul. 2025. [Online]. Available: <https://research.trychroma.com/context-rot>
- [3] O. Bianchi *et al.*, “Lost in the haystack: Smaller needles are more difficult for LLMs to find,” arXiv:2505.18148v1, May 2025. [Online]. Available: <https://arxiv.org/abs/2505.18148v1>
- [4] M. Li *et al.*, “Haystack Engineering: Context engineering for heterogeneous and agentic long-context evaluation,” arXiv:2510.07414, 2025. [Online]. Available: <https://doi.org/10.48550/arXiv.2510.07414>
- [5] U. Shaham, M. Ivgi, A. Efrat, J. Berant, and O. Levy, “ZeroSCROLLS: A zero-shot benchmark for long text understanding,” in *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023, pp. 7977–7989.
- [6] C. An *et al.*, “L-Eval: Instituting standardized evaluation for long context language models,” in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (ACL 2024)*, Vol. 1: *Long Papers*, 2024, pp. 14388–14411.
- [7] Z. Dong, T. Tang, J. Li, W. X. Zhao, and J.-R. Wen, “BAMBOO: A comprehensive benchmark for evaluating long text modeling capacities of large language models,” in *Proc. Joint Int. Conf. Comput. Linguistics, Lang. Resour. Eval. (LREC-COLING 2024)*, 2024, pp. 2086–2099.
- [8] J. Li, M. Wang, Z. Zheng, and M. Zhang, “LooGLE: Can long-context language models understand long contexts?” in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (ACL 2024)*, Vol. 1: *Long Papers*, 2024, pp. 16304–16333.
- [9] X. Zhang *et al.*, “Bench: Extending long context evaluation beyond 100K tokens,” in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (ACL 2024)*, Vol. 1: *Long Papers*, 2024, pp. 15262–15277.
- [10] C.-P. Hsieh *et al.*, “RULER: What’s the real context size of your long-context language models?” arXiv:2404.06654, 2024. [Online]. Available: <https://doi.org/10.48550/arXiv.2404.06654>
- [11] H. Yen *et al.*, “HELMET: How to evaluate long-context language models effectively and thoroughly,” arXiv:2410.02694, 2024. [Online]. Available: <https://doi.org/10.48550/arXiv.2410.02694>
- [12] M. Li, S. Zhang, Y. Liu, and K. Chen, “NeedleBench: Can LLMs do retrieval and reasoning in 1 million context window?” arXiv:2407.11963v1, 2024. [Online]. Available: <https://arxiv.org/abs/2407.11963v1>
- [13] G. Zhang *et al.*, “Path-of-Thoughts: Extracting and following paths for robust relational reasoning with large language models,” arXiv:2412.17963v1, Dec. 2024. [Online]. Available: <https://arxiv.org/abs/2412.17963v1>
- [14] L. Hilgert, D. Liu, and J. Niehues, “Evaluating and training long-context large language models for question answering on scientific papers,” in *Proc. 1st Workshop Customizable NLP (CustomNLP4U)*, 2024, pp. 220–236.
- [15] V. Tripathi, U. Allu, and B. Ahmed, “The instruction gap: LLMs get lost in following instruction,” arXiv:2601.03269v1, Dec. 2025. [Online]. Available: <https://doi.org/10.48550/arXiv.2601.03269>